LoadBond

GRAPHIQUES INNOVATION CHALLENGE · FINANCE & FINTECH

Know what your security actually secures.

Clean-energy projects post millions in letters of credit to keep their grid connection. LoadBond reads the filed agreement, links every number to its source sentence, and computes the most that security can become, including when the honest answer is “at least” or “unknown”.

Working prototype · tested on a real FERC-accepted PJM agreement

THE PROBLEM · SCALE

The grid connection queue is where clean energy waits, and waiting costs collateral.

2,061 GW

of generation and storage actively seeking interconnection at the end of 2025, across 8,244 projects.

549 GW

already hold a draft or executed interconnection agreement but are not yet operating.

61 mo

median time from interconnection request to operation for projects completed in 2025 (22 months in 2008).

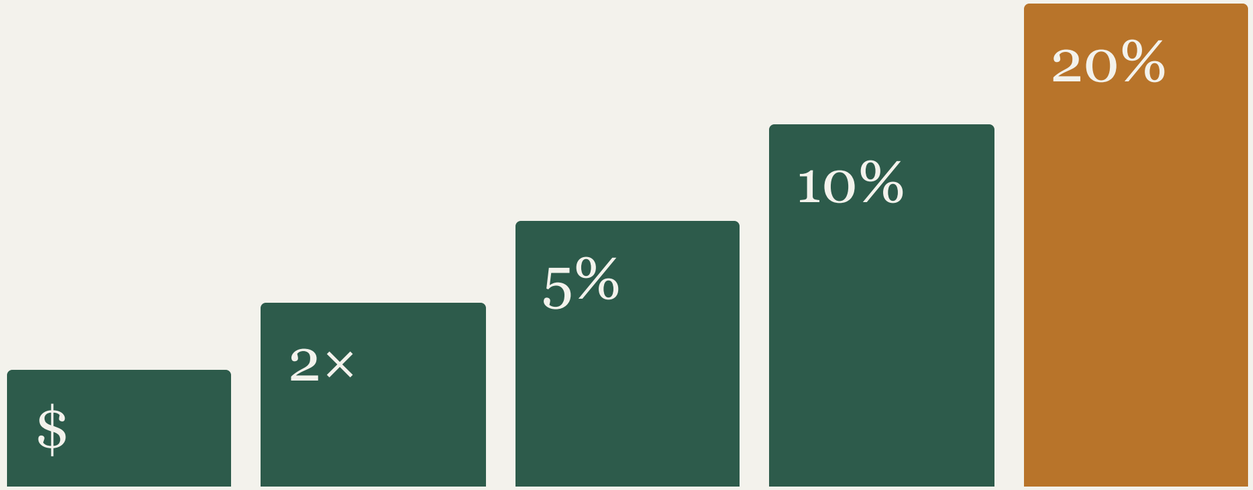
13%

of capacity requested 2000–2020 was operating by 2025. 75% withdrew.

Every one of those projects must post financial security at each stage, and keep it posted for years while amounts change with milestones, scope, tax status and withdrawals.

Source: Lawrence Berkeley National Laboratory, “Queued Up: 2026 Edition” (data through end of 2025), emp.lbl.gov/queues

FERC Order 2023 made security escalate at every step.



Study deposit: $35k–$250k by project size

Readiness deposit: twice the study deposit

of network-upgrade cost at restudy

of network-upgrade cost at facilities study

of network-upgrade cost when the agreement is signed

Sources: FERC, “Explainer on the Interconnection Final Rule” (Order No. 2023, July 2023); Orrick summary of deposit schedule (Aug 2023); FERC Order 2023-A explainer (cash, letters of credit or surety bonds)

- Posted as cash, a bank letter of credit, or (since Order 2023-A) a surety bond.
- Withdrawal penalties apply if leaving the queue harms other projects.
- Letters of credit cost about **1–3% of face per year** and use up the developer’s bank credit lines.
- One developer’s portfolio averaged **$0.05 per watt** of security across interconnection, PPAs, purchase orders and swaps, ranging up to $0.37/W.

Sources: Segue Infrastructure, “The Cash Cow That No American Bank Seems To Want” and “Deposits, Deposits, Deposits…”; Evergreen Surety, “Posting Collateral…” (a surety provider; reports 100 bp rising to 200–275 bp)

A REAL AGREEMENT

One question a treasury team must answer: what is the most this security can become?

Chestnut Solar, 74.9 MW, Virginia. Interconnection Service Agreement with PJM and Dominion Energy Virginia, accepted by FERC (Docket ER17-1701) in 2017. Public filing, 163 pages.

WHERE	WHAT IT SAYS
Page 14, §5.0	Security "in the amount of $2,111,149"
Appendix 2, §24.4	More security required if tax safe-harbor documentation is missing. **Amount not stated.**
§11.2.1	Reduced as work completes; changed with scope. **Amounts not stated.**
PJM Tariff §212.4	When it can be drawn or returned. **A different document.**

A SPREADSHEET SAYS

~~$2,111,149~~

THE CORRECT ANSWER

At least $2,111,149

because §24.4 can raise it by an amount the agreement never states.

Source: pjm.com/pjmfiles/pub/planning/project-queues/isa/aa2_165_isa.pdf · SHA-256 562a469f…

WHY CURRENT APPROACHES FAIL

Contracts are conditional. The tools that track them are not.

Spreadsheets hold one number

A cell can say $2,111,149. It cannot say “unless documentation is missing” or “until an unstated reduction applies”, so a false maximum looks exactly like a real one.

Contract tools store, not compute

Document repositories find the PDF. They do not combine releases, step-ups, expiries and draw conditions into the exposure that can actually exist on a date.

AI summaries guess

A chatbot will happily turn an unstated amount into a number. In collateral, a confident wrong figure is worse than “unknown”: it either strands capital or hides a draw.

The cost of being wrong runs both ways: **overstating** keeps credit lines tied up at 1–3% a year; **understating** means a surprise demand for more security or a draw.

Our characterization of current practice is a hypothesis from public sources and has not yet been validated with treasury teams (see slide 11).

THE SOLUTION

From a 163-page filing to a defensible number, every figure traced to its sentence.

01 · UPLOAD

Load the filed agreement

PDFs up to 500 pages; pages without text are flagged, not silently skipped.

02 · FIND

Local AI proposes

A small model on the user's own machine flags sentences with money and conditions. Nothing leaves the computer.

03 · REVIEW

A person approves exact quotes

Nothing counts until a named reviewer accepts the verbatim source sentence.

04 · ENUMERATE

Every feasible state

A rules engine checks every combination of events and conditions the contract allows, exactly.

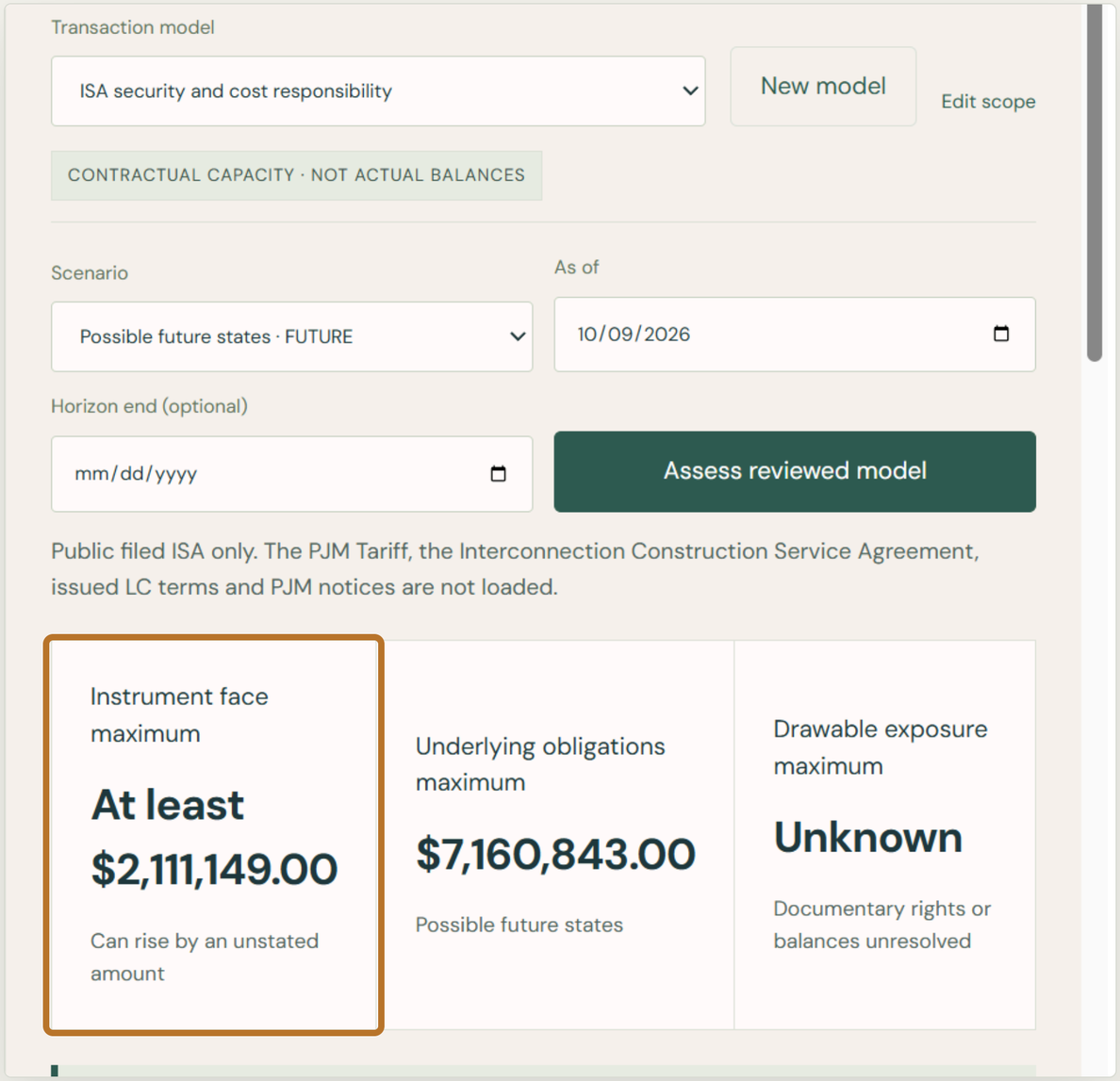
05 · DECIDE

A brief you can replay

The maximum, the state that produces it, the sources, and a frozen snapshot that recomputes identically.

FOUR SEPARATE ANSWERS

Instrument face · amount that can be drawn · underlying obligations · reimbursement debt. Merging them is how exposure gets miscounted.



Three answers. Only one is a plain number.

- **At least $2,111,149** for the letter of credit, with a finding naming §24.4 as the reason.
- **$7,160,843** in cost obligations, the two estimates in Specifications §4.1 and §4.2.
- **Unknown** drawable amount, because draw terms live in the PJM Tariff, which is not loaded. LoadBond says so instead of guessing zero.

32 feasible contract states checked for May 2017 to Dec 2018. Re-running the saved snapshot reproduces the result byte for byte.

Screenshot of the running prototype · samples/public-isa/trial-result.json

INNOVATION & UNIQUENESS

Built to be wrong out loud, never quietly.

Unknown stays unknown

Missing facts, unreviewed claims and unloaded documents block completion or are shown as unknown. They never default to zero.

Lower bounds, not false maximums

When a clause can raise security by an unstated amount, the answer becomes “at least”. This rule came directly from testing the real agreement.

Exact, not simulated

Every feasible state is enumerated, not sampled. The maximum comes with the exact combination of events that produces it.

AI finds, people decide

The local model only proposes source sentences. Calculation uses reviewed records alone, so a model error cannot reach the number unseen.

Together these make LoadBond an **auditable calculator for conditional collateral**: a category between document storage and generic AI, where the output can be defended to a bank, an auditor or a utility.

QUALITY OF RESEARCH · WHAT TESTING FOUND

We tested on a real filing and published the failures.

REAL-AGREEMENT TRIAL FOUND	STATUS
163-page filing rejected by upload limits	Fixed
Reported “maximum” was silently too low	Fixed: “at least”
74.9 MW rejected as not a whole number	Fixed
Local AI timed out on a long filing	Fixed: 162/162 chunks
No way to search 163 pages	Fixed: find in document
Security terms sit in a separate tariff	Open

LOCAL AI, SCORED ONCE ON UNSEEN DOCUMENTS	RESULT
Real amounts it found (recall)	100%
Proposals that were right (precision)	60%
Contract clauses: precision / recall	88% / 83%
Real filing: found the $2,111,149	Yes
Real filing: told totals from components	No

Verdict: the AI is a good finder of money in long documents, not a reliable source of structured data. That is why review is mandatory, and why the engine, not the model, produces the number.

85 automated tests · held-out sets were never used for tuning · full write-up in docs/public-agreement-trial.md

POTENTIAL IMPACT & BUSINESS MODEL

Collateral that sits idle is capital not building clean energy.

$21k–$63k

yearly letter-of-credit fees on Chestnut's $2.11M security alone, at 1–3% of face.

Arithmetic on sourced fee range

~$27B

order-of-magnitude illustration: 549 GW with agreements × $0.05/W average security.

Combines LBNL and Segue figures; not a measured market size

75%

of queued capacity historically withdraws. Each exit needs a clear answer on what security returns and what is owed.

LBNL Queued Up 2026

First customer: treasury and finance teams at mid-size renewable and storage developers, with several projects in queues and limited bank credit lines.

Model (hypothesis): annual subscription per project under management, priced below one month of fees on the security it tracks; counsel and lenders as referral partners.

FEASIBILITY · ROADMAP · HONEST LIMITS

It runs today. Here is what must be true next.

BUILT

- Working web app with a Python rules engine and audit trail
- Local AI on a laptop (4-billion-parameter model, free)
- Tested end to end on a 163-page FERC filing

NEXT 90 DAYS

- Interview 10 developer treasury teams; reconstruct one real security history each
- Load agreement + tariff as one package
- Mark estimates subject to true-up; stop double-counting totals

NOT YET TRUE

- No customers, banks or utilities have validated the value
- Not legal advice; one real agreement tested
- AI extraction is experimental and below our own quality gate

Stop rule. If treasury teams cannot show a real case where conditional security was misjudged, we stop investing in this thesis. The test is designed to be failable.

LoadBond

IN ONE LINE

“At least $2,111,149” is the honest answer. LoadBond is built to give it.

Exact exposure for the security that keeps clean-energy projects in the queue, traced to the source, with every unknown left visible.

Prototype built with AI coding assistance; started from an earlier synthetic prototype and research report (disclosed)